# Exit Ticket · reflect and retrieve

## Exit Ticket

One hexadecimal digit represents four binary bits.

## Use key vocabulary from memory

1. Explain Hexadecimal in your own words.

2. Explain Nibble in your own words.

3. Explain Logical shift in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.



RHS COMPUTING / OCR H446 / LESSON 003 DRAFT · TEACHER REVIEW

# Do Now · retrieval

## Hexadecimal, shifts & masks

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

Choose a type

### Q2

Read an unsigned byte

### Q3

Encode a negative number

### Q4

Add and check the range

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Hexadecimal links directly to binary — Each hexadecimal digit maps to exactly four bits because four bits have 16 possible patterns. — CODE:
$6D_{16}$
6 → 0110
D → 1101
= $01101101_2$

## Core idea 2

Logical shifts move bits — A logical shift inserts zeros into the empty positions. Bits leaving the fixed width are lost.

## Core idea 3

AND and OR masks — AND: mask 1 preserves a bit; mask 0 clears it. — OR: mask 1 sets a bit; mask 0 preserves it.

## Annotate the key visual

Explain which labels or features show the main process.





# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 2

Control an eight-bit register
Bits are numbered 7…0 from left to right. Give an operation and an eight-bit mask for each independent task. Preserve all bits not mentioned.
Clear bits 7 and 6. [2]
Set bits 3 and 0. [2]
Toggle bits 5 and 2. [2]
Original exam-style application; one mark for the operation and one for the mask.

Check that you have used the technical terms and explained your decisions.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Hexadecimal → binary

Write the binary equivalent of hexadecimal B7E .

Paraphrased exam question; original value and mark retained. OCR H446/01, page 17 .

Check that you have used the technical terms and explained your decisions.





# New Learning · applying the ideas

## Core idea 4

XOR toggles selected bits — XOR produces 1 when the two input bits differ. A mask bit of 1 flips the corresponding data bit; 0 leaves it unchanged.

## Vocabulary · use the correct technical terms

Hexadecimal — Base 16, using digits 0–9 and letters A–F.

Nibble — A group of four bits.

Logical shift — Moving bits left or right, filling empty positions with zeros.

Mask — A bit pattern combined with another using a bitwise operation.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

## Problem

Link denary, hexadecimal and binary

## Follow the worked method

1. Find the hex digits — CODE:
173 ÷ 16 = 10 remainder 13 — The quotient gives the first hex digit; the remainder gives the second.

2. Use four bits per hex digit — CODE:
A → 1010
D → 1101

$AD_{16} = 10101101_2$

3. Return to denary — CODE:
(10 × 16) + 13 = 173

128 + 32 + 8 + 4 + 1 = 173 — For binary → hex, group bits from the right in fours and pad the left group with zeros if needed.

## Why this method works

Identify the decision, rule or invariant that makes the method correct.





# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Build the answer

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.